

# VANELLE AYONTA NDJOUTSE

Data Scientist • AI & ML Engineer • NLP Engineer

Statistician-Economist Engineer (ISE) – Intelligent decision-support systems

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## PROFILE

Statistician-Economist Engineer (ISSEA) and **AI / Data Engineer** building **intelligent decision-support systems** end to end. I combine rigorous statistical and econometric expertise with strong command of **Machine Learning, Deep Learning** and **LLMs** to turn complex data – textual and numerical alike – into reliable decision levers. Experience gained inside a **central bank** (BEAC) and the **National Institute of Statistics**, where I shipped robust, secure and production-ready **agentic RAG, NLP** and document-automation pipelines. Autonomous and impact-driven, I transform raw data into clear, actionable insight.

## PROFESSIONAL EXPERIENCE

**AI & Data Science Engineer** (Internship) Mar. 2026 – May 2026

Yaoundé, CM

BEAC – Bank of Central African States • Financial Inclusion & Innovation Dept.

- › Designed and deployed a complete **agentic RAG pipeline** – scraping of **22 international sources**, optimised chunking, **768D** embeddings, **FAISS** vector index, LLM agents and automated export (Word / PDF / e-mail) – fully automating the department's information watch.
- › Orchestrated a **multi-LLM** architecture with **Pydantic-AI**, secured by **guardrails** and **prompt-injection** protection.
- › Built a **10-page Streamlit** interface and Dockerised the solution for reproducible, production-ready deployment.
- › Automated extraction from **scanned PDFs** via **Claude API**, eliminating a time-consuming manual process.
- › Contributed to the redesign of the survey questionnaire and specifications of the **CEMAC** financial-inclusion data-collection platform.

**Data Analyst** (Internship) Aug. 2025 – Oct. 2025

Yaoundé, CM

INS – National Institute of Statistics • Business Statistics Dept.

- › Designed **statistical analysis plans** for national data-collection operations.
- › Processed and analysed structured survey databases and mined unstructured sources (reports, official documents) to produce **decision-ready syntheses**.

**Mathematics Teacher & Department Head** Nov. 2018 – Oct. 2023

Melong, CM

Melong Bilingual High School

- › Coordinated the mathematics department's teaching activities and **led a team of teachers**.
- › Drove learning assessment and academic planning over **5 years**.

## PROJECTS & RESEARCH

**NLP Analysis of Cameroon's Finance Laws** 2026

Measure the semantic alignment of the 2026 Finance Law with the National Development Strategy (SND 30) objectives and assess budget compliance.

Stack : Python · sentence-transformers · CamemBERT · semantic similarity

→ Compliance-analysis pipeline over a legislative corpus, informing public-policy evaluation.

## Deep-Learning Quality Control

2025–2026

Build a binary quality classifier from **28 features** and **2,148 samples**.

Stack : PyTorch · MLP · Optuna · Streamlit

→ Model tuned via hyperparameter search (Optuna) with an **interactive prediction dashboard**.

## Geomarketing & Commercial Strategy

Optimise market entry and territorial coverage across banking, telecoms, mobile money, insurance and agribusiness in Cameroon.

Stack : Spatial analysis · segmentation · data visualization

→ Recommendations on bank-branch placement, telecom sales-force sizing and agribusiness territorial defence.

## Data Collection & Delivery Automation

Make field collection and data access more reliable through custom web tools.

Stack : HTML · Google Sheets · Netlify · Claude API · Streamlit

→ Deployed a custom web questionnaire and an app for automatic data extraction from scanned PDFs.

## TECHNICAL SKILLS

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|                                 |                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------|
| <b>Programming</b>              | Python · R · SQL · Bash                                                                    |
| <b>Machine Learning</b>         | scikit-learn · SVM · predictive modelling · supervised & unsupervised learning             |
| <b>Deep Learning</b>            | PyTorch · neural networks (MLP) · Optuna                                                   |
| <b>LLM &amp; GenAI</b>          | Agentic RAG · Pydantic-AI · Claude API (Anthropic) · guardrails · prompt-injection defence |
| <b>NLP</b>                      | CamemBERT · sentence-transformers · Hugging Face · FAISS · semantic similarity             |
| <b>MLOps &amp; Deployment</b>   | Docker · Git / GitHub · GitHub Actions · Streamlit · Netlify                               |
| <b>Data &amp; Visualization</b> | Power BI · Metabase · Pandas · data storytelling                                           |
| <b>Databases</b>                | MongoDB · MySQL                                                                            |
| <b>Statistics &amp; Econ.</b>   | Stata · SPSS · econometrics · sampling design                                              |
| <b>Data Collection</b>          | Survey Solutions · KoboToolbox · CPro · Google Forms                                       |
| <b>Soft skills</b>              | Analytical rigour · autonomy · leadership · communication to non-technical audiences       |

## EDUCATION

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|--------------------------------------------------------------------|-------------|
| <b>Statistician-Economist Engineer Degree (ISE)</b>                | 2023 – 2026 |
| ISSEA — Sub-regional Institute of Statistics and Applied Economics | Yaoundé, CM |
| <b>DIPES I — Mathematics (Teaching Diploma)</b>                    | 2015 – 2018 |
| ENS — Higher Teacher Training College of Yaoundé                   | Yaoundé, CM |
| <b>B.Sc. in Applied Mathematics</b>                                | 2015 – 2019 |
| University of Yaoundé I                                            | Yaoundé, CM |

## LEADERSHIP & ENGAGEMENT

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### Vice-President, Student Council

Apr. 2025 – Apr. 2026

- › Led student projects and mediated between the student body and the administration.

**Cluster Leader – Teacher Training Program**

Oct. 2019 – Oct. 2023

*AIMS – African Institute for Mathematical Sciences*

Melong, CM

- › Facilitated training groups, mentored peers and coordinated scientific communication.

**CERTIFICATIONS & TRAINING**

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- › Monitoring & Evaluation of a development project.
- › Designing projects respectful of the environment and local populations.
- › Decentralisation: challenges and processes.

**LANGUAGES**

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**French**

Native

**English**

Professional working (B2)

*References available upon request.*